Recap: Game Theory

- basic components of a game
- Prisoners' Dilemma
- (Strictly) dominated strategies
- Bank Runs (no dominated strategies)

depositor 2

depositor 1	withdraw	don't
withdraw	25, 25	50, 0
don't	0, 50	110, 110

two Nash Equilibria (withdraw, withdraw) and (don't, don't)

General Method of Finding Nash Equilibria

1) consider each of player 2's strategies and circle player 1's highest payoff in each column (if two are the same → circle both)

2) repeat for player 2, consider each of player 1's strategies and circle 2's highest payoff in each row

3) any cell with both payoffs circled is a Nash Equilibrium

ex.

		2		
		D	E	F
	A	4, 2	13, 6	1, 3
1	B	3, 10	0, 0	15, 2
	C	12, 14	4, 11	5, 4

two Nash Equilibria: (C, D) and (A, E)

Classic Games

1) Battle of the Sexes

		Jill: Inn	Jill: Pub
Jack	Inn	1, 2*	0, 0
	Pub	0, 0	2, 1*

again, two Nash Equilibria
→ coordination game

applications: Compatibility standards
social media
social goods

Notes: how do we end up with one equilibrium or the other? → "nudges" or social conventions "focal points"

how do you win? → Committing to an action

convincing your rival that you are committed to a strategy can be a big advantage

2) Game of Chicken

		Cal: straight	Cal: swerve
Ricky	straight	-2, -2	1, -1*
	swerve	-1, 1*	0, 0

two Nash Equilibria (straight, swerve) and (swerve, straight)
→ anti-coordination game

applications: brinkmanship, war, labour negotiations

Similar lesson: convince your rival that you are serious and committed to an action is often enough to win

3) The Stag Hunt (Rosseau)

		②	
		stag	hare
①	stag	2, 2*	0, 1
	hare	1, 0	1, 1*

each player can hunt hare on their own, but hunting stag requires cooperation

similar to the Prisoners' Dilemma <u>but</u> two Nash Equilibria

tradeoff between safety and cooperation

applications: evolutionary biology, theories of trust and social structures